

Top Design Patterns for Interviews — Expanded Country & Company Edition

Frequency Ratings + Company-Level Data — Saudi Arabia | Dubai/UAE | Malaysia | India Tier-2 | India Tier-1/60LPA+

Interview Prep Reference · Updated July 2026

Top Design Patterns Asked in Interviews

Expanded Edition — Country-Wise Company Data, Difficulty Levels & Pattern Demand

How to Use This Document

This is the expanded version of your original frequency guide. It keeps the core pattern-frequency and pattern-recognition material, and adds a company-level research layer: for each of your five target markets (Saudi Arabia, Dubai/UAE, Malaysia, India Tier-2, and India Tier-1/60LPA+), you now have a table of roughly 25–30 employers with what's actually known about their LLD/design-pattern interview rounds — pulled from Glassdoor, LeetCode Discuss, GeeksforGeeks, Blind, and company engineering write-ups.

India Tier-2 vs. Tier-1/60LPA+ — these are deliberately separate markets, not a relabeling. Tier-2 covers well-funded product companies and fintechs (Razorpay, PhonePe, Swiggy, CRED-tier). Tier-1/60LPA+ covers the FAANG-adjacent and top-comp bracket (Amazon India, Google India, Flipkart, Atlassian Bengaluru, Rippling, Goldman Sachs India-tier) — a measurably higher bar, confirmed via a dedicated research pass rather than assumed.

A note on data honesty before you use this: public interview-report volume varies enormously by company. Big regional players (Careem, Talabat, Property Finder, Maybank, Razorpay, Swiggy) have dozens of dated, specific candidate reports. Many smaller or younger companies (especially in Saudi Arabia and Malaysia) have close to zero public interview write-ups. Every row below is labeled **Confirmed** (a real, sourced candidate report exists) or **Inferred** (no direct report found — pattern guess is based on the company's domain, e.g. a payments company probably tests Strategy/Adapter for gateway routing). Treat Inferred rows as reasonable priors, not facts. Nothing below is a fabricated quote or invented source.

Rating scale: Very High · High · Medium · Low

1. Master Frequency Table

#	Pattern	Category	Global Frequency	Notably explicit in	Where It Shows Up
1	Strategy	Behavioral	Very High	All 5 markets — the single most universal pattern, incl. India Tier-1 (Microsoft cache eviction, VMware)	Pricing/discount engines, payment gateways, sorting/eviction policies
2	Factory / Abstract Factory	Creational	Very High	UAE (Property Finder), India	Parking Lot, Vehicle systems, Notification channels
3	Singleton	Creational	Very High	Malaysia (asked to be written live at Maybank & AirAsia)	Logger, Config Manager, DB Connection Pool
4	Observer	Behavioral	Very High		

#	Pattern	Category	Global Frequency	Notably explicit in	Where It Shows Up
				India (PhonePe pub-sub), UAE (Yellow.ai pub-sub)	Notification systems, stock tickers, event-driven/pub-sub systems
5	Decorator	Structural	High	General — no strong regional skew found	Pizza/Coffee ordering, middleware chains, UI customization
6	Builder	Creational	High	India (MakeMyTrip — thread-safe Builder w/ atomics)	Complex object construction, HTTP request builder, query builder
7	State	Behavioral	High	India (order-lifecycle heavy: Swiggy, Zomato, Meesho, Nykaa)	Elevator, Vending Machine, Order lifecycle
8	Adapter	Structural	Medium	UAE/Saudi (open banking, 3rd-party gateway integration)	Third-party API integration, legacy system wrapping
9	Command	Behavioral	Medium	Malaysia (Gojek — ride/order request objects)	Undo/Redo, Task Scheduler, Remote Control
10	Chain of Responsibility	Behavioral	Medium	Saudi/UAE fintech (fraud/approval pipelines)	Middleware/filters, approval workflows, logging levels
11	Composite	Structural	Medium	General	File System design, Org hierarchy, UI trees
12	Facade	Structural	Low–Medium	General	Simplifying subsystem access
13	Proxy	Structural	Low–Medium	India (Razorpay Rate Limiter)	Caching layer, Rate limiter, Lazy loading
14	Template Method	Behavioral	Low	General	Report generation, algorithm skeletons
15	Prototype	Creational	Low	General	Object cloning, invoice/template duplication

Prep priority unchanged: master rows 1–7 in depth. Rows 8–11 need working knowledge. Rows 12–15 need definition + one example.

2. Classic LLD Problems → Patterns Tested

LLD Problem	Patterns Tested
Parking Lot System	Factory, Strategy, Singleton
Elevator System	State, Strategy, Observer
Splitwise / Expense Sharing	Strategy, Observer
Movie Ticket Booking (BookMyShow-style)	Factory, Singleton, State, Strategy
Food Delivery (Swiggy/Zomato/Talabat-style)	Observer, Strategy, State, Factory
Ride-Sharing (Uber/Careem-style)	Strategy, Observer, Factory, State
Rate Limiter	Strategy, Proxy, Singleton
Notification System (Email/SMS/Push)	Observer, Factory, Strategy
Vending Machine	State, Strategy
Chess / Tic-Tac-Toe	State, Strategy, Command
Logging Framework	Singleton, Chain of Responsibility, Strategy
LRU Cache Design	Strategy (eviction policy), Singleton
ATM Machine	State, Strategy, Command

LLD Problem	Patterns Tested
Library Management System	Factory, Observer, Singleton
Hotel Booking System	Strategy, State, Factory, Observer

2b. Company-Reported LLD Problems (real, dated candidate reports, 2024-2025)

These are problems specifically confirmed via candidate write-ups during this research pass — a good "surprise-proof" list to add to your practice set:

Problem	Reported at
Class design for a Twitter-like object model	Careem (Dubai/Riyadh)
Yelp-like nearby-places system (HLD)	Careem
Parking Lot with role-based discounts (UML, extensibility)	Property Finder (Dubai)
SAGA pattern / distributed transaction orchestration	Property Finder
Pub-sub event-management service	Yellow.ai (Dubai)
Multi-tenant SaaS architecture	Emirates NBD (Dubai)
Message queue / pub-sub with regex-based subscribers	PhonePe (India)
Multilevel cache design	PhonePe
Splitwise-style expense manager (with partial splits)	Swiggy, Groww, ShareChat (India)
Restaurant/food-ordering order-lifecycle system	Zomato (India)
Thread-safe Builder pattern (atomics)	MakeMyTrip (India)
Rate Limiter (Token Bucket / Sliding Window)	Razorpay (India)
Payment Ledger, Loan EMI Calculator, Logger system	Razorpay
LRU Cache / Key-Value Store	PolicyBazaar, CRED (India)
Doctor-appointment / clinic booking system	Meesho (India)
Facebook/Twitter-like feed with follow + pagination	Meesho
Car rental system	Nykaa, CARS24 (India)
Snake & Ladder (modular, demoable)	CARS24
Audit Log System / File-parsing service (Strategy)	Freshworks (India)
Reverse-proxy with round-robin load balancing	BrowserStack (India)
URL Shortener	iFlix (Malaysia)
Cache with custom eviction policy	Shopee (Malaysia/SEA hub)
"Design an online cart system"	Carsome (Malaysia)
Microservices vs. monolith trade-off discussion	HungerStation (Saudi)
Highly-available backend for millions of sensor events/day	Aramco Digital (Saudi)

3. Recognition Triggers (Pattern-Spotting Cheat Sheet)

If the problem says...	Think...
"Multiple ways to calculate/process something (payment, discount, route)"	Strategy
"Object creation should be centralized/hidden from client"	Factory
"Only one instance should ever exist (config, logger, connection pool)"	Singleton
"Notify multiple parts of the system when something changes"	Observer
"Add optional features/behavior without changing the base class"	Decorator
"Object has many optional fields / complex step-by-step construction"	Builder
"Behavior changes based on internal status/lifecycle"	State
"Integrate with a third-party API that has a different interface"	Adapter
"Need undo, redo, or queued/deferred actions"	Command
"Request passes through multiple handlers (validation, auth, logging)"	Chain of Responsibility
"Tree-like structure — folders/files, org chart, menus"	Composite
"Simplify a complex subsystem behind one clean interface"	Facade
"Control/restrict access to an object (caching, lazy-load, permissions)"	Proxy
"Object creation is expensive and you need many similar copies"	Prototype
"Same algorithm skeleton, different steps per subclass"	Template Method

4. Country-Wise Deep Dive

Each section below: what dominates, difficulty, interview format, then a company table. Confidence key — **C** = Confirmed via a real sourced report, **I** = Inferred (no direct report found, domain-based guess).

Saudi Arabia

Most-asked pattern: Strategy and Observer, tied — both largely *implied* by domain (payment/pricing logic, order/delivery notifications) rather than named outright. **SOLID principles and "justify your architectural choice" reasoning** are tested more explicitly than GoF pattern names at most companies.

Difficulty: Trails global big-tech LLD bars — most regional companies rate 2.7–3.1/5 on Glassdoor (Medium). Exceptions run Hard: Amazon, Uber, and Aramco Digital (rigorous system-design grilling on caching, Kafka, high-availability).

Format: Typically 3–4 rounds over 2–7 weeks. A single combined system-design/LLD round is more common than a dedicated LLD-only round, except at Careem (explicit architecture-and-patterns round) and HungerStation (explicit "architecture and design patterns" questions). Recurring candidate complaint across multiple companies: slow feedback / ghosting after technical rounds.

Company	Sector	Conf.	Difficulty	Top patterns tested	Common LLD problems / topics	Source
Careem	Ride-hailing/super-app	C	Medium–Hard	Explicit design-pattern selection & justification	Twitter-like class design (LLD); Yelp-like nearby-places (HLD)	InterviewQuery , Blind
Talabat (Delivery Hero KSA)	Food delivery	C	Medium (2.7/5)	Design patterns named in coding round; DDD	TDD kata + refactoring; domain-driven system design	Glassdoor
HungerStation (stc-owned)	Food delivery	C	Medium (2.9/5)	Explicit "architecture and design patterns"; SOLID	Microservices vs. monolith; DB indexing/query efficiency	Glassdoor
Tamara	Fintech (BNPL)	C	Medium (3/5)	Strategy/State (inferred from domain)	Build "SplitPayment" from UI mock (1hr); integrating new project with legacy monolith	Glassdoor
Tabby	Fintech (BNPL)	C (process) / I (patterns)	Easy–Medium	Strategy, State (inferred)		Glassdoor

Company	Sector	Conf.	Difficulty	Top patterns tested	Common LLD problems / topics	Source
					System design round; code review exercise; ACID/transaction isolation	
noon (KSA ops)	E-commerce	C (process) / I (patterns)	Medium (3.1/5)	Not itemized	DSA (LeetCode-Medium) + DB design	Glassdoor
stc / STC Group	Telecom/digital	C	Medium (2.87/5)	MVP, MVVM explicitly named (Android roles)	OOP theory, coroutines/multithreading, mini-project	general synthesis
Aramco Digital	Energy-tech	C	Hard	No GoF names, but rigorous "why this choice" grilling	HA backend for millions of sensor events/day; cache-failure scenarios	Medium
Mozn	AI/RegTech	C	Medium (2.9/5)	Not confirmed	BFS-based DSA + open-ended system design	Glassdoor
Amazon (Riyadh)	Big Tech/Cloud	C (global loop)	Hard	Full GoF toolkit, SOLID	Design LRU Cache, Parking Lot, Unix "find" command	codezym.com/lld/amazon
Uber (regional)	Ride-hailing	C (global loop)	Hard	Concurrency-aware design	Key-value store handling millions of req + TTL	Blind
Lean Technologies	Fintech infra (open banking)	C (process) / I (content)	Medium	Adapter/Strategy (inferred)	4-stage process + case study to VP Finance	Glassdoor
Unifonic	CPaaS/messaging	C (process) / I (content)	Medium (3.08/5)	Observer (inferred)	1hr technical interview on past projects	Glassdoor
Al Rajhi Bank / Neoleap	Digital banking	C (process) / I (content)	Medium	State (transaction lifecycle), Strategy (fee rules) — inferred	Panel case study, then individual interviews	Glassdoor
SNB / SAB digital	Banking	C (process) / I (content)	Unknown	Strategy/State (inferred)	3-round process	Glassdoor
Foodics	Retail-tech/POS SaaS	C (process) / I (content)	Unknown	Observer, State, Strategy (inferred)	HR → take-home → tech interview → 2nd take-home	Glassdoor
Geidea	Fintech/payments	C (difficulty only)	Hard (8/10)	Strategy/Adapter/CoR (inferred)	Not surfaced	Indeed
Elm Company	Gov-tech (PIF-owned)	C (process) / I (content)	Medium	Not confirmed	2-phase, mostly behavioral	Glassdoor
Accenture (KSA centers)	IT consulting	C (global)	Medium	Factory, Strategy, Observer, Singleton flagged as "most important"	Scenario-based pattern implementation; HashMap/TreeMap internals	InterviewQuery
TCS (KSA delivery centers)	IT services	C (global)	Easy-Medium	OOP fundamentals; polymorphism a "favorite"	OOP + DSA + SQL rounds	GeeksforGeeks
Cognizant (KSA)	IT services	C (global)	Easy-Medium	OOP fundamentals	Technical → Managerial → HR	general synthesis
Mrsool	On-demand delivery	C (process) / I (content)	Medium-Hard	Not confirmed	Phone screen → algo round → system design	Glassdoor
Jahez	Food delivery	I	Unknown	State, Observer, Strategy (inferred)	No public data found	general regional inference
Salla	E-commerce SaaS	I	Unknown	Factory, Strategy, Adapter (inferred)	Indeed confirms no interview data exists	Indeed
Sary	B2B marketplace	I	Unknown	Strategy, Observer (inferred)	Indeed confirms no interview data exists	Indeed
Nana	Quick commerce	I	Unknown	Strategy, Observer, Decorator (inferred)	No public data found	general regional inference
Zid	E-commerce SaaS	I	Unknown	Factory, Strategy, Adapter (inferred)	No public data found	general regional inference
Rabbit	Quick commerce (new KSA entrant)	I	Unknown	Strategy, Observer (inferred)	Too new to market, no data	general regional inference
PayTabs	Fintech/payments	I	Unknown	Strategy, Adapter, CoR (inferred)	Stack confirmed as Java/Spring only	general regional inference

Dubai / UAE

Most-asked pattern: Strategy and Factory lead, but the region's real distinguishing feature is **SAGA / distributed-transaction patterns and pub-sub design** showing up explicitly (Property Finder, Yellow.ai) — more architecture-of-microservices flavored than pure GoF-pattern-naming.

Difficulty: Clusters Medium region-wide (2.7–3.0/5 on Glassdoor), below FAANG. Property Finder is the standout exception — candidates describe a genuinely hard, multi-round LLD+HLD+DSA day.

Format: Consistent pipeline across confirmed companies: recruiter screen → DSA round (often LeetCode-Medium) → dedicated component/class-design or LLD round → HLD round → hiring-manager/behavioral. Property Finder and Careem are the two companies that most explicitly label a round "Low Level Design."

Company	Sector	Conf.	Difficulty	Top patterns tested	Common LLD problems / topics	Source
Careem (incl. NOW/Pay)	Ride-hailing/super-app	C	Medium–Hard	OOD class design, bar-raiser eval	Class design (social/Twitter-like); Yelp-like HLD	Glassdoor
Property Finder	Proptech	C — strongest UAE signal	Hard	Explicit "Low Level Design" round; SAGA; microservice orchestration	Parking Lot UML (role-based discounts, extensibility); ride-hailing HLD	LeetCode Discuss , Glassdoor
Talabat (Delivery Hero)	Food delivery	C	Medium	TDD, DDD, queuing, observability	TDD kata/refactoring; scalability/caching/RPC system design	Medium , Glassdoor
Dubizzle / Bayut	Classifieds/property	C	Easy–Medium (2.94/5)	OOP, caching strategies, queuing, security patterns (senior roles)	ER diagram of a social-media-like app + queries	Glassdoor
Amazon (Dubai/MENA)	E-commerce/cloud	C (global loop)	Hard	SOLID, OOD, bar-raiser	Amazon's standard LLD bank (Parking Lot, Library-style)	Amazon interview prep
Emirates NBD	Banking	C (partial)	Medium	Multitenancy architecture; MVVM (iOS roles)	Multi-tenant SaaS architecture question	Medium
ADCB	Banking	C (generic)	Medium	OOD principles, distributed systems	Java, networking (TCP/SSL), no specific LLD problem documented	Glassdoor
Etisalat / e&	Telecom	C (partial)	Medium (2.7/5)	Java, Hibernate, Spring, DB design	No specific LLD case documented	Glassdoor
Emirates Group (airline IT)	Aviation/IT	C (partial)	Medium–Hard	Microservice architecture discussion	"Describe architecture of your project/microservice components"	Glassdoor
Yellow.ai (has Dubai ops)	Conversational AI	C (company-wide)	Medium–Hard	Explicit pub-sub LLD round	Event-management service design, chatbot HLD, pub-sub LLD	Glassdoor
Deliveroo (owns InstaShop UAE)	Food delivery	C (global process)	Medium–Hard	Concurrency, scalability, locking	Architecture round on concurrency/high-scale/locking	Deliveroo careers guide
Zoho (Dubai presence)	SaaS	C (global) / I (UAE-specific)	Medium–Hard	Hands-on OOP pattern implementation	Mini-app/CLI tool demonstrating OOP patterns	InterviewQuery
G42	AI/tech holding	I	Medium (2.73/5)	Infra-heavy (Linux, Terraform, K8s, AWS) more than OOD	No LLD report found	Glassdoor
Mashreq (Neo)	Digital banking	I	Medium	Not confirmed	Generic technical rounds only	Glassdoor
Noon.com	E-commerce	I	Medium	Strategy/Factory/Observer (inferred)	Product catalog, cart, pricing (inferred)	general regional inference
Kitopi	Cloud-kitchen tech	I	Medium	Order/kitchen orchestration (inferred)	No SE-specific report found	Glassdoor
Network International	Fintech/payments	I	Medium–Hard	Strategy/Adapter for gateway integration (inferred)	Payment processing system (inferred)	general regional inference
du (EITC)	Telecom	I	Medium	Not confirmed	No LLD-specific report found	general regional inference
Presight AI (G42)	AI/data analytics	I	Medium	SQL/Python/data-modeling focus (found); no LLD	Not confirmed	general regional inference
Chalhoub Group (tech arm)	Luxury retail digital	I	Medium	Not confirmed	Generic technical questions only	Glassdoor
InstaShop (Deliveroo-owned)	Grocery/q-commerce	I	Medium	Not confirmed	No SE-specific questions found	Glassdoor

Company	Sector	Conf.	Difficulty	Top patterns tested	Common LLD problems / topics	Source
Trukker	Logistics tech	I	Medium	Not confirmed	No data found	general regional inference
Fetchr	Last-mile logistics	I	Medium	Not confirmed	Company has scaled down significantly	general regional inference
YallaCompare	Fintech comparison	I	Medium	Not confirmed	No data found	general regional inference
ENOC / Petrolina	Energy/retail tech	I	Easy-Medium	Not confirmed	Generic HR-style questions only	Glassdoor
Dubai REST / Ejari	Gov proptech	I	Medium	Not confirmed	No candidate reports found	general regional inference
Bayzat	InsurTech/HRTech	I	Medium	Not confirmed	No data found	general regional inference
Anghami	Music streaming (Abu Dhabi HQ)	I	Medium	Not confirmed	No data found	general regional inference
Tabby / Tamara (UAE hiring)	Fintech/BNPL	I	Medium-Hard	Strategy/State (inferred)	Payment/checkout system design (inferred)	general regional inference

Note: Zomato exited the UAE market in 2019 (sold to Talabat/Delivery Hero) — not included as a current independent employer.

Malaysia

Most-asked pattern: Singleton is uniquely explicit here — candidates at both Maybank and AirAsia report being asked to write it out live (double-checked locking discussed at Maybank). Strategy is a close second, especially at e-commerce/superapp employers.

Difficulty: Medium at Malaysia-local companies (banks, StoreHub, Carsome). One notch harder — approaching Hard/near-FAANG — at Singapore-HQ'd regional offices hiring into KL (Shopee, Agoda, Grab, Gojek), so candidates targeting SEA-regional tech hubs should prep at the higher bar even if based in Malaysia.

Format: Combined HLD+LLD or "system design + code review" rounds are more common than a pure whiteboard LLD round — Agoda explicitly critiques pattern usage during a code review, Gojek asks candidates to discuss patterns in code they just wrote, Maybank runs two back-to-back system-design problems. Live coding (Spring Boot CRUD, Singleton implementation) is frequently expected, not just diagrams.

Company	Sector	Conf.	Difficulty	Top patterns tested	Common LLD problems / topics	Source
Maybank	Digital banking	C	Medium-Hard	Singleton (double-checked locking, explicit)	Layered architecture, Hibernate N+1, 2 system-design problems onsite	GeeksforGeeks , Glassdoor
AirAsia / Capital A	Airline/Superapp (MOVE)	C	Medium-Hard	Singleton (asked to write live)	CAP theorem, SOLID, scaling, O(n²)→O(n) refinement, Spring Boot CRUD	Glassdoor
Shopee (SEA hub incl. MY)	E-commerce	C (regional)	Hard	Strategy, Factory, Observer, Singleton	Cache with eviction policy; how trees are stored on disk; DB/network fundamentals	Glassdoor
Agoda (large KL presence)	Travel-tech	C (regional)	Medium-Hard	Strategy (explicitly named in code review)	LLD Notification System; "Booking API" code review, spot bugs/propose patterns	Prepfully
Carsome	Automotive e-commerce	C	Medium	Strategy/Factory implied	Design online cart system; composite-key DB management	Glassdoor
GoTo / Gojek	Ride-hailing/Superapp	C (regional)	Hard	Command (ride/order requests), Strategy (matching)	Machine coding → project discussion → HLD+LLD on recent project	GitHub , Blind
JPMorgan (KL Payment Tech hub)	Banking/Fintech	C (active KL hiring)	Hard	SOLID-driven design discussion	Core Java + data engineering + design patterns; fault-tolerant architecture	educative
MoneyLion (KL eng. center)	Fintech	C	Medium	Not pattern-named	SQL vs NoSQL trade-off, algorithm round	Glassdoor
CIMB / CIMB Octo	Digital banking	C (general CIMB)	Medium	Layered/service architecture (inferred)	HackerRank + easy DSA; backend stack justification	Glassdoor

Company	Sector	Conf.	Difficulty	Top patterns tested	Common LLD problems / topics	Source
iFlix	Streaming/OTT	C	Medium	Cloud/scalability focus over classic OOD	Design a URL Shortener (onsite); subscription mgmt take-home	Glassdoor
Grab (MY eng. hub)	Ride-hailing/Superapp	I	Hard	Strategy, Factory, Observer, State (industry-standard)	Parking-lot/ride-matching-style LLD (peer-superapp norm)	general regional inference
Curlec (Razorpay)	Fintech/Payments	I (parent-co data)	Medium-Hard	Strategy, Observer	In-memory pub-sub, photo-sharing app, severity-based logger (parent co.)	general inference from Razorpay reports
Boost (Axiata)	Fintech/eWallet	I	Medium	Strategy/Factory (wallet/txn norm)	Not confirmed	Glassdoor listing
PayNet / DuitNow	Payment infrastructure	I	Medium-Hard	Strategy, Chain of Responsibility (routing/validation)	Not found — domain-based reasoning (payment switch design)	general regional inference
Touch 'n Go eWallet	Fintech/eWallet	I	Medium	Strategy, Observer (wallet/txn norm)	Not confirmed	general regional inference
StoreHub	Retail/POS SaaS	I	Medium	Not confirmed	80 MY reviews found, none SWE-technical	Glassdoor
PropertyGuru	Proptech	I	Medium	Not confirmed	No data surfaced	general regional inference
Fave	Fintech/rewards	I	Unknown	Not confirmed	No data surfaced	general regional inference
ServisHero	Services marketplace	I	Unknown	Not confirmed	No data surfaced	general regional inference
Aeon Credit	Consumer finance	I	Medium	Not confirmed	No data surfaced	general regional inference
Public Bank (digital)	Banking	I	Medium	Not confirmed	No data surfaced	general regional inference
RHB	Banking	I	Medium	Not confirmed	No data surfaced	general regional inference
IBM Malaysia	Enterprise IT	I (global data)	Medium	Strategy, Factory, Observer, Singleton (global)	Vending Machine, Parking Lot System (global bank)	general inference
Dell Technologies (Cyberjaya/ Penang)	Hardware/enterprise IT	I (global data)	Medium	OOP/pattern review cited generally	System design discussion (senior roles)	general inference
Micron Malaysia (Penang)	Semiconductor	I (global data)	Medium	OOP principles, not pattern-specific	DSA + embedded systems + some system design	general inference
Intel Malaysia (Penang)	Semiconductor	I (global data)	Hard (when LLD appears)	Not pattern-named; C++/memory-management skew	Parking Lot, Traffic Signal Controller, Elevator (global bank)	general inference
HSBC / Standard Chartered / PayPal / Visa / Mastercard (MY)	Banking/fintech caps	I	Medium-Hard	Not confirmed	Not confirmed — search budget exhausted before company-specific data pulled	general regional inference
Accenture Malaysia	IT consulting	I	Medium	Not confirmed	Varies heavily by client account	general regional inference

India Tier-2

Most-asked pattern: Strategy is the clear #1 — it demonstrates polymorphism/composition cleanly and shows up everywhere. **Observer, Factory, Singleton** round out the top 4, and **State** is unusually heavy here specifically because of the volume of order/booking-lifecycle problems (Swiggy, Zomato, Meesho, Nykaa, CARS24).

Difficulty: Easy-Medium at the classic IT-services majors (TCS, Infosys, Wipro, Cognizant, Capgemini, HCLTech, LTIMindtree — LLD is often folded into general OOP/project discussion rather than a dedicated timed round). Medium-Hard at well-funded product companies and fintechs (CRED, Razorpay, Freshworks, PhonePe, MakeMyTrip, Postman, BrowserStack) — comparable to India Tier-1 (Flipkart/Myntra-caliber) rigor, generally a notch below FAANG's design depth.

Format: A 90–150 minute standalone "Machine Coding" round producing working, compilable code plus a verbal design-defense, sometimes preceded by a separate DSA/LLD-theory round. **India-specific trend:** interviewers here more often ask candidates to directly *name and justify* a specific design pattern (Postman's process is a documented example) — a more definitional/theory-adjacent style than typical US interviews for the same companies.

Company	Sector	Conf.	Difficulty	Top patterns tested	Common LLD problems / topics	Source
Razorpay	Fintech/Payments	C	Medium-Hard	Factory, Singleton, Strategy		LeetCode , Dataford

Company	Sector	Conf.	Difficulty	Top patterns tested	Common LLD problems / topics	Source
					Rate Limiter (Token Bucket/Sliding Window), Payment Ledger, Expense Splitter, Loan EMI Calculator, Logger	
PhonePe	Fintech	C	Medium-Hard	Observer (pub-sub), Strategy	Message-queueing/pub-sub system, multilevel cache design, support-ticketing system	LeetCode , enginebogie
CRED	Fintech	C	Hard	SOLID-heavy, no single forced pattern	LRU Cache, Key-Value Store, Payment Recommendation system	Medium
Swiggy	Food delivery/Q-commerce	C	Medium	Strategy, State, Observer	Splitwise-style expense manager; vaccine booking system	Glassdoor
Zomato	Food delivery	C	Medium	Observer, State, Strategy	Restaurant/order-lifecycle system, delivery assignment	Medium
Meesho	E-commerce/Social commerce	C	Medium-Hard	Strategy, State	Doctor-appointment booking; Twitter-like feed w/ follow+pagination; inventory-blocking system	LeetCode
MakeMyTrip	Travel/OTA	C	Medium-Hard	Builder (thread-safe), Strategy	Vending Machine; thread-safe Builder w/ atomics; Uber-like driver allocation	LeetCode
Freshworks	SaaS/Enterprise	C	Medium-Hard	Strategy	File-parsing service (Strategy), Audit Log System, LRU Cache; 4-round pipeline	GeeksforGeeks
Postman	SaaS/DevTools	C	Medium-Hard	Strategy, Singleton, Factory, Observer	LLD for caching service; explicit note that patterns are named/justified directly	Glassdoor
BrowserStack	SaaS/DevTools	C	Medium-Hard	Strategy, State	3-problem round (2 DSA + 1 LLD); reverse-proxy w/ round-robin load balancing	Medium
Nykaa	E-commerce/Beauty	C	Medium-Hard	Strategy, State	Car-rental system; delivery-system design; multi-outlet w/ microservices	LeetCode
CARS24	Used-vehicle marketplace	C	Medium	Strategy, State	Vehicle/car-rental LLD; Snake & Ladder (90-min modular round)	GeeksforGeeks
Groww	Fintech/Wealthtech	C	Medium	Strategy, State	Splitwise-style app; stock-exchange bid-matching; Spotify-like music app	LeetCode
ShareChat	Social media/Content	C	Medium	Singleton, Factory, Strategy	Expense-splitting system w/ partial splits	Glassdoor
PolicyBazaar	Insurtech/Fintech	C	Easy-Medium	Caching-related design	LRU Cache as first-round LLD problem	LeetCode
Zoho	SaaS/Enterprise	C	Medium	OOP-first, less pattern-name-driven	Pattern-based pseudo-code problems, DB schema for feature extension	GeeksforGeeks
Innovaccer	Healthtech SaaS	C	Medium	Strategy, layered DAO/service design	SD-III: DSA+LLD round 1, HLD+DB round 2	GeeksforGeeks
Delhivery	Logistics	C (generic)	Medium	Strategy, Factory	3 rounds/4 problems spanning DSA, system design, LLD	Naukri Code360
Sprinklr India	SaaS/Martech	C	Medium (3.22/5)	Component/composition design	Build a bar-chart component from scratch (machine coding)	Glassdoor
Dream11	Fantasy sports/Gaming	C (generic)	Medium	Strategy, State, Factory, Observer	LLD round confirmed focused on OOD/patterns/testable code	Glassdoor

Company	Sector	Conf.	Difficulty	Top patterns tested	Common LLD problems / topics	Source
Rapido	Mobility (bike-taxi)	C (weak)	Medium	Strategy, Factory	Discussion of candidate's own past project design	Naukri Code360
Paytm	Fintech	C (generic)	Medium	Strategy, Factory	Listed among companies where LLD/machine coding is a mandatory round	Medium
Ola	Mobility/Ride-hailing	C (generic)	Medium	Singleton, Factory, Strategy, Observer	Named as machine-coding-round employer; vehicle/dispatch modeling implied	general pattern-guide reference
Infosys	IT Services	C (generic)	Easy–Medium	Strategy, Observer, Factory, Singleton	Java/Spring Boot fundamentals; OOP applied to projects	Glassdoor
Cognizant	IT Services	C (generic)	Easy–Medium	Strategy, Observer, Factory, Singleton	OOPs tested in AI-Bot round + final technical round	TechPrep
HCLTech	IT Services	C (generic, 2.7/5)	Easy	OOP principles applied to projects	Core Java/Spring Boot/React/SQL/REST; 1–2 live coding problems	Glassdoor
LTIMindtree	IT Services	I	Easy–Medium	Strategy, Factory	String/DP problems; LLD/pair-programming round increasingly common	GeeksforGeeks
TCS Digital	IT Services	I	Easy–Medium	Strategy, Factory, Singleton	Basic OOP/SOLID discussion in digital-hire technical rounds	general regional inference
Wipro	IT Services	I	Easy–Medium	SOLID principles, OOP basics	Core OOP, DBMS normalization, OS; scalable API/caching for experienced hires	TechPrep
Capgemini	IT Services	I	Easy–Medium	Strategy, Factory	OOP concepts (encapsulation/inheritance/abstraction)	general regional inference
Urban Company	Home-services marketplace	I	Medium	Strategy, Observer (booking/dispatch)	Service-provider/booking assignment (sector norm)	general regional inference
InMobi	Adtech	I	Medium	Strategy, Observer	Ad-serving/bidding-engine style LLD plausible	general regional inference
Lenskart	E-commerce/Eyewear	I	Medium	Strategy, Factory	Inventory/order-management LLD plausible (omnichannel retail)	general regional inference
Darwinbox	HRTech SaaS	I	Medium	Strategy, State	Campus round observed = MCQ+DSA, not explicit LLD	LeetCode
upGrad	Edtech	I	Easy–Medium	Strategy, Factory	Course/LMS entity-modeling LLD plausible	general regional inference

India Tier-1 / 60LPA+

Most-asked pattern: **Strategy** again leads (Microsoft's cache-eviction LLD, VMware, and the general prep-guide consensus all converge here), with **Singleton, Factory/Builder, and Observer** close behind — the same top group as Tier-2, but tested with meaningfully more rigor. **SOLID-principle articulation is a baseline expectation, not a bonus**, especially at Goldman Sachs, Atlassian, Rippling, and Walmart Global Tech.

Difficulty: Clearly exceeds India Tier-2. Where Tier-2 LLD rounds are generally Medium–Hard and pattern-naming-focused, this tier (Amazon, Google, Uber, Atlassian, Rippling, Microsoft, Goldman Sachs) more often demands working, compiling code with concurrency/thread-safety follow-ups (mutexes, thread pools, thread-safe eviction) layered on top of the base design. Multiple sources explicitly flag that candidates fail not on the core design but on unhandled edge cases or designs that require a large rewrite when a scale-up twist is introduced mid-interview (Atlassian, Rippling).

Format: A distinguishing structural trait at this tier is the **bar-raiser/veto mechanic** — Amazon's Bar Raiser and Microsoft's "As Appropriate" round both carry outsized, near-veto authority over the hire decision, and are frequently where the hardest LLD variant surfaces (Amazon's multi-lot Parking Lot twist was specifically reported in a Bar Raiser round). Banking/fintech players in this tier (Goldman Sachs, Morgan Stanley, Visa, PayPal, Mastercard) consistently run a **dedicated, separately-graded OOD/LLD round distinct from DSA** — a structural formality less commonly seen as a standalone gate at Tier-2 companies. Coverage confidence varies sharply by company: firms with large, English-heavy candidate-blogging communities (Amazon, Uber, Rippling, Atlassian, Walmart, Myntra) have rich, specific, recent (2024–2025) evidence; Meta India, Cisco, Grab, and Deutsche Bank yielded only generic or non-India-specific material and are marked Inferred.

Company	Sector	Conf.	Difficulty	Level/Comp context	Top patterns tested	Common LLD problems / topics	Source
Amazon India	E-commerce/Cloud	C	Hard	SDE2/SDE3(L5)/L4, Bar Raiser has veto power	Strategy, general OOD/SOLID	Parking Lot (multi-lot variant), meeting/room reservation, Twitter API design, string+concurrency LLD	LeetCode , GeeksforGeeks
Google India	Search/Cloud	C	Hard	L4/L5/L6 (L6 bar notably higher: multi-DC, partitioning)	Iterator, general OOD	Library/Iterator coding; LLD folded into system design in final ~10 min; tagging systems	Blind (L6) , LeetCode (L4 India)
Flipkart	E-commerce	C	Medium-Hard	SDE2	Strategy, State-machine patterns	Multi-level cache (L1/L2/L3), Wallet system, scalable Tic-Tac-Toe (nXn, multi-user), FSM, autocomplete	GeeksforGeeks
Walmart Global Tech (India)	Retail-tech	C	Medium-Hard	SDE3/Senior SE; Staff Engineer runs HLD round	Strategy, Dependency Injection, SOLID	Restaurant Reservation & Serving System; Aarogya-Setu-like app HLD	GeeksforGeeks (SDE3) , dev.to
Uber India	Ride-hailing	C	Hard	SDE2/L4	Singleton, Strategy, multithreading	Train-platform scheduling w/ waiting queue; "Design Facebook" w/ 3 working features + concurrency; Linux file-system (mkdir/cd/lis)	Medium , Blind
Adobe India	Software/Creative-cloud	C	Medium	Senior SE (system design weighted more than DSA at senior levels)	General OOD principles	Booking app (order creation→execution flow); CoderPad 45-60 min format	LinkedIn , interviewing.io
Salesforce India	SaaS/CRM	C	Medium-Hard	SMTS (Senior Member Technical Staff)	General class/interface design	Bike Rental app (searchBikes w/ filters); Notification service (LLD+HLD combined)	LeetCode , Medium
Microsoft India (IDC)	Cloud/Enterprise	C	Hard	Senior SDE / L63 (comp ~65L-1.1Cr per candidate report)	Strategy (eviction policy)	LRU Cache design (OOD+SOLID+Strategy); thread-safe transaction filter (mutex/locks)	Medium , Rehearsal AI
Atlassian Bengaluru	DevTools/SaaS	C	Hard	SSE/Principal Engineer levels	SOLID-driven extensibility focus	Color Picker Tool, Tag Management System, Confluence-like feed, Web Crawler, Snake Game; graded on "scale-up requires minimal change"	Medium , jointaro (Principal SE)
Goldman Sachs India	Investment Bank	C	Medium-Hard	SDE2/Java Developer, dedicated 60-min OOD round	SOLID-principles based OOD	Dedicated "Low-Level Design" round testing OOD/OOPs/clean code + thread pools/synchronization	Medium
Morgan Stanley India	Investment Bank	C	Medium	2.5+ yrs experienced / Senior	General OOP/polymorphism	Online banking management system design w/ concurrency consideration	GitHub gist , GeeksforGeeks
Directi / Media.net	Adtech	C	Medium-Hard	SDE/SRE	Singleton, Factory, Strategy, Adapter, Chain of Responsibility, Decorator, Template, Observer (explicitly named)	Socket-programming-based systems; LLD with deliberately unwritten/discussed requirements	GeeksforGeeks , Medium
Meta India	Social/Ads	I (weak)	Hard	E5/E6 — India presence mostly non-core-SDE; loop generic	Java OOD encouraged given LLD-round possibility	ML system design dominates over classic LLD in found evidence	Hellointerview (E5) , general inference
ServiceNow India	Enterprise SaaS	C	Medium-Hard	Staff SE / IC3, SSE	General class-modeling	Loan Management System (EMI calc); employee-hierarchy DB design	LeetCode (SSE IC3)

Company	Sector	Conf.	Difficulty	Level/Comp context	Top patterns tested	Common LLD problems / topics	Source
Oracle India	Enterprise/Cloud	C	Medium	Software Developer (4.5 YOE), OCI roles	Strategy, Builder, Singleton, Observer (explicitly named)	Movie-booking system HLD w/ concurrent transactions & scalability	GeeksforGeeks , LeetCode (OCI)
Visa India	Fintech/Payments	C	Hard	Senior/Staff SWE (10 YOE example)	Concurrency-aware OOD	BookMyShow-type LLD w/ concurrent booking handling + DB design	Medium (Staff SE, 10 YOE)
Mastercard India	Fintech/Payments	C (light)	Medium	SDE1 (evidence skews junior)	Singleton, Factory, Builder (explicit, e.g. Burger-Shop Builder)	Car system design w/ patterns	LeetCode
PayPal India	Fintech/Payments	C	Hard	SE3/T24, SDE2/L2	Cloud-agnostic pattern selection (AWS/Azure abstraction)	File-based processing system w/ status tracking (HLD+LLD combined), class diagrams	Substack (SE3) , Medium (Senior SE)
Cisco India	Networking/Infra	I	Medium	General inference — no India-specific candidate account surfaced	Token bucket/sliding window style	Rate Limiter design (from general prep guides, not a verified transcript)	general inference
VMware India	Virtualization/Infra	C (light)	Medium	Senior Java Developer	Strategy (noted most common in India LLD rounds generally)	LRU Cache design; linked-list manipulation in coding round	Medium (Senior Java Dev)
Nutanix India	Cloud Infra/HCI	C	Medium–Hard	MTS-3/MTS-4 (senior IC track)	UML-driven class design	Cross-browser Bookmark Manager, Paint application (Windows Paint clone)	LeetCode (MTS-3) , LeetCode (MTS-4)
Myntra	E-commerce/Fashion	C	Medium–Hard	SDE1/SDE2/SSE	Domain-driven class modeling	Shopping Cart w/ discount logic, Fashion Recommendation, Inventory atomic reservation, Flash Sale HLD (Redis DECR)	Medium (SSE) , GeeksforGeeks (SDE2)
Cars24	Auto/Marketplace	C	Medium	SDE2, Senior Java Developer	Extensible OOD	Vehicle rental system (APIs, DB schema, ER diagram + working code)	LeetCode (Senior Java Dev SDE2)
Ather Energy	EV/Hardware-tech	C (light)	Medium	SDE/Backend Engineer	Not explicitly named	API Rate Limiter design in coding round (borderline LLD/ coding hybrid)	Medium
Rippling India	HR-tech/SaaS (US-HQ)	C (strong)	Hard	Senior SE/SDE2 — "extremely selective, fails vast majority"	SOLID, extensibility-first OOD	Rule-Based System design; employee-record grouping/ aggregation (extend to arbitrary key)	jointaro , enginebogie
Grab (India-facing eng)	Ride-hailing/Fintech (SG-HQ)	I	Medium–Hard	SDE2 (general APAC loop, no India-specific transcript)	Singleton, Factory, Strategy, Observer (general list, not verified)	Working module w/ compiling code, DS + design-pattern judged	general inference
Deutsche Bank India	Investment Bank	I	Medium	Inferred from peer banks (GS/ MS pattern)	SOLID, general OOD (sector norm)	No direct evidence found; likely similar to GS/MS banking-domain LLD (loan/trade/ portfolio systems)	general inference

5. Cross-Country Comparison

Country	#1 Pattern	#2 Pattern	Typical Difficulty	Distinctive trait
Saudi Arabia	Strategy (implied)	Observer (implied)	Medium, Hard at global cos	"Justify your architecture" reasoning valued over naming patterns

Country	#1 Pattern	#2 Pattern	Typical Difficulty	Distinctive trait
Dubai/UAE	Strategy / Factory	Observer / SAGA-pub-sub	Medium, Hard at Property Finder/Amazon	Explicit "Low Level Design"-labeled rounds; SAGA/distributed-transaction focus
Malaysia	Singleton (uniquely explicit)	Strategy	Medium, Hard at SG-linked hubs (Shopee/Grab/Agoda)	Live-code a pattern (not just diagram it); concurrency follow-ups common
India Tier-2	Strategy	Observer / State	Easy-Medium at IT-services, Medium-Hard at product cos	Interviewers ask you to <i>name and justify</i> the pattern directly
India Tier-1/60LPA+	Strategy	Singleton / Factory-Builder	Medium-Hard baseline, Hard at Amazon/Google/Atlassian/Rippling	Bar-raiser/veto rounds; concurrency + scale-up follow-ups layered on top of the base design

Bottom line for your prep: Strategy is non-negotiable everywhere. If you're interviewing in Malaysia, drill Singleton (including thread-safety/double-checked locking) hard enough to write it from memory. If you're interviewing in UAE, add SAGA and pub-sub/event-driven system design to your HLD practice, not just class diagrams. India Tier-2 interviewers are the most likely to ask you to say the pattern name out loud and defend it — practice articulating "I used X instead of Y because Z" for your top 7 patterns. If you're targeting India Tier-1/60LPA+, the design itself is table stakes — what separates offers from rejections is handling a live scale-up or concurrency twist without a large rewrite, so rehearse extending your design under a follow-up, not just producing it once.

6. How Interviewers Actually Test This (Across All Regions)

Requirements → Identify entities → Class diagram → Code → Pattern justification

The pattern name is rarely the point — **the ability to justify why Strategy fits here instead of an if-else chain** is what's actually being scored. A strong answer always includes the trade-off: *"I used Strategy instead of a switch statement because new payment methods will be added frequently, and this keeps the Order class closed for modification."*

7. Quick Self-Check Before Any LLD Round

- [] Can I state Strategy, Factory, Singleton, Observer definitions in under 15 seconds each?
- [] Can I sketch Parking Lot + Elevator + Notification System class diagrams from memory?
- [] Can I write Singleton with thread-safety (double-checked locking) from memory — critical for Malaysia?
- [] Can I design a SAGA/distributed-transaction flow at a high level — relevant for UAE?
- [] For each pattern, do I have **one real production example**, not just a textbook one?
- [] Can I explain why I'd choose Strategy over a simple if-else, out loud, unprompted?
- [] Can I take my design and extend it live under a scale-up or concurrency twist without a large rewrite — critical for India Tier-1/60LPA+ bar-raiser rounds?

8. Methodology & Confidence Notes

This edition was built by researching ~25–30 companies per market (Saudi Arabia, UAE, Malaysia, India Tier-2, India Tier-1/60LPA+) across Glassdoor, LeetCode Discuss, GeeksforGeeks, Blind/TeamBlind, and Medium engineering write-ups. A few honest limits to keep in mind:

- Coverage is uneven.** India Tier-1/60LPA+ and Tier-2, plus Malaysia's regional-hub companies (Shopee, Agoda, Grab), have the richest public data — large, English-heavy candidate-blogging communities (Amazon, Uber, Rippling, Atlassian, Walmart, Myntra) mean specific, recent (2024–2025) evidence. Saudi Arabia and several younger Malaysian/UAE startups have thin-to-nonexistent public interview reports — those rows are marked **Inferred** and should be treated as reasonable priors based on company domain, not fact.
- "Confirmed" ≠ "verbatim current."** Sourced reports are dated candidate write-ups (mostly 2023–2025); interview formats change over time, especially at fast-growing startups.
- No fabricated data.** Every Inferred row states plainly that no direct report was found — none of those rows cite a fake source or invented quote.
- India Tier-2 vs. Tier-1/60LPA+ is a deliberate split, not a relabeling** — different company sets, researched separately, with a measurably different interview bar (see the note at the top of this document).
- Total companies profiled: **28 (Saudi) + 29 (UAE) + 28 (Malaysia) + 33 (India Tier-2) + 27 (India Tier-1/60LPA+) ≈ 145 companies** across the five markets.

Companion reference: see the dedicated Prototype Pattern deep-dive guide for full-depth treatment (production examples, shallow vs deep copy, Q&A bank) — apply the same depth to Strategy, Factory, Observer, and Singleton next, since those are your Very High priority patterns across every target market. The pattern-deep-dive template (MASTER-PROMPT-TEMPLATE.md) now cross-links its Market Calibration section directly to this document's data — check there before assuming a pattern's interview weight in a given market.